Year 10 Higher

REVISION PACK

Progress Check 2

Thursday 5 November 2026

Name: _____

Class: _____

Units in this pack

- Graphs 1
- Number 2
- Shape and Space 1

Graphs 1 · Gradients and equations of lines (Lessons 3.1, 3.3)

Worked example: a missing coordinate

A straight line passes through the points (1, 4) and (5, k). The gradient of the line is 2. Work out the value of k.

Step 1 Gradient formula	$\frac{k-4}{5-1}=2$
Step 2 Simplify the bottom	$\frac{k-4}{4}=2$
Step 3 Multiply by 4	$k-4=8$
Step 4 Add 4	$k=12$

Practice: a missing coordinate

1.1 A straight line passes through the points (3, 2) and $(7, k)$. The gradient of the line is 4. Work out the value of k.

1.2 A straight line passes through the points (1, 9) and $(4, k)$. The gradient of the line is −2. Work out the value of k.

1.3 A straight line passes through the points (−2, 5) and $(3, k)$. The gradient of the line is 3. Work out the value of k.

1.4 A straight line passes through the points (0, −3) and $(4, k)$. The gradient of the line is 2. Work out the value of k.

1.5 A straight line passes through the points (2, 1) and $(5, k)$. The gradient of the line is −3. Work out the value of k.

1.6 A straight line passes through the points (−1, 6) and $(3, k)$. The gradient of the line is 2. Work out the value of k.

Graphs 1 · Gradients and equations of lines (Lessons 3.1, 3.3)

Worked example: the equation of a line through two points

Find an equation of the straight line that passes through the points (−2, 9) and (2, −3).

Step 1	Gradient	$m = \dfrac{-3-9}{2-(-2)} = -3$
Step 2	Put in a point	$9 = -3 \times (-2) + c$
Step 3	Solve	$c = 3$
Step 4	Answer	$y = -3x + 3$

Practice: the equation of a line through two points

1.7 Find an equation of the straight line that passes through the points (2, 7) and (5, 19).

1.8 Find an equation of the straight line that passes through the points (−1, −1) and (3, 11).

1.9 Find an equation of the straight line that passes through the points (−1, 7) and (2, −2).

1.10 Find an equation of the straight line that passes through the points (1, 6) and (4, −6).

1.11 Find an equation of the straight line that passes through the points (−3, 10) and (1, 2).

1.12 Find an equation of the straight line that passes through the points (0, −4) and (2, 6).

Number 2 · Percentages (Lesson 4.1)

Worked example: percentage loss

The table shows what Sara paid for a car and what Sara sold it for. Work out Sara's percentage loss.

Bought for	Sold for
KD 2500	KD 2100

Step 1	Loss	$2500 - 2100 = 400$
Step 2	Loss ÷ cost × 100	$\frac{400}{2500} \times 100 = 16\%$
Step 3	Answer	16%

Practice: percentage loss

2.1 The table shows what Ali paid for a car and what Ali sold it for. Work out Ali's percentage loss.

Bought for	Sold for
KD 18 000	KD 14 940

2.2 The table shows what Mona paid for a sofa and what Mona sold it for. Work out Mona's percentage loss.

Bought for	Sold for
KD 640	KD 528

2.3 The table shows what Omar paid for a boat and what Omar sold it for. Work out Omar's percentage loss.

Bought for	Sold for
KD 2500	KD 2150

2.4 The table shows what Huda paid for a car and what Huda sold it for. Work out Huda's percentage loss.

Bought for	Sold for
KD 12 000	KD 9000

2.5 The table shows what Fahad paid for a TV and what Fahad sold it for. Work out Fahad's percentage loss.

Bought for	Sold for
KD 450	KD 396

2.6 The table shows what Laila paid for a car and what Laila sold it for. Work out Laila's percentage loss.

Bought for	Sold for
KD 8000	KD 7400

Number 2 · Percentages (Lesson 4.1)

Worked example: reverse percentages

The value of a car decreases by 15% each year. After one year it is worth KD 5100. Work out the value of the car when new.

Step 1	After one year	$100\% - 15\% = 85\%$
Step 2	Multiplier	0.85
Step 3	Divide	$5100 \div 0.85 = 6000$
Step 4	Answer	$\text{KD } 6000$

Practice: reverse percentages

2.7 The value of a car decreases by 24% each year. After one year it is worth KD 5320. Work out the value of a car when new.

2.8 The value of a motorbike decreases by 15% each year. After one year it is worth KD 3060. Work out the value of a motorbike when new.

2.9 The value of a laptop decreases by 8% each year. After one year it is worth KD 391. Work out the value of a laptop when new.

2.10 The value of a phone decreases by 15% each year. After one year it is worth KD 204. Work out the value of a phone when new.

2.11 The value of a boat decreases by 5% each year. After one year it is worth KD 2850. Work out the value of a boat when new.

2.12 The value of a van decreases by 20% each year. After one year it is worth KD 8320. Work out the value of a van when new.

Number 2 · Recurring decimals (Lesson 4.2)

Worked example: recurring decimal proofs

Prove algebraically that $0.1\dot{6} = \frac{1}{6}$

Step 1	Let x be the decimal	$x = 0.1666\ldots$
Step 2	Multiply by 100	$100x = 16.666\ldots$
Step 3	Multiply by 10	$10x = 1.666\ldots$
Step 4	Subtract	$90x = 15$
Step 5	Answer	$x = \frac{15}{90} = \frac{1}{6}$

Practice: recurring decimal proofs

3.1 Prove algebraically that $0.2\dot{7} = \frac{5}{18}$

3.2 Prove algebraically that $0.3\dot{8} = \frac{7}{18}$

3.3 Prove algebraically that $0.4\dot{1} = \frac{37}{90}$

3.4 Prove algebraically that $0.1\dot{5} = \frac{7}{45}$

3.5 Prove algebraically that $0.2\dot{5} = \frac{23}{90}$

3.6 Prove algebraically that $0.5\dot{2} = \frac{47}{90}$

Shape and Space 1 · Angles (Lessons 5.1, 5.3)

Worked example: the exterior angle of a triangle

In triangle ABC, the side BC is extended to D. Angle $BAC = (x + 10)^\circ$, angle $ABC = (2x + 20)^\circ$ and angle $ACD = (4x - 5)^\circ$. Work out the size of angle ACD.

Step 1	Exterior = two opposite angles	$x + 10 + 2x + 20 = 4x - 5$
Step 2	Simplify	$3x + 30 = 4x - 5$
Step 3	Solve	$x = 35$
Step 4	Angle ACD	$4 \times 35 - 5 = 135^\circ$

Practice: the exterior angle of a triangle

4.1

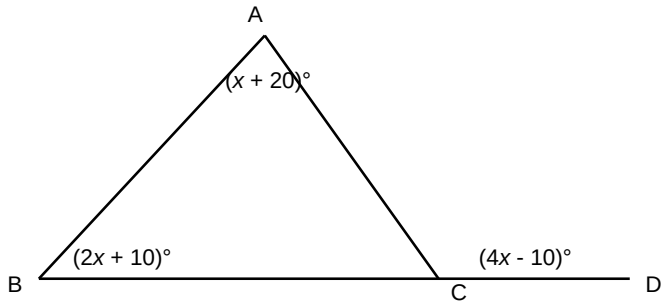


Diagram **NOT** accurately drawn

The diagram shows triangle ABC with the side BC extended to D. Work out the size of angle ACD.

4.2

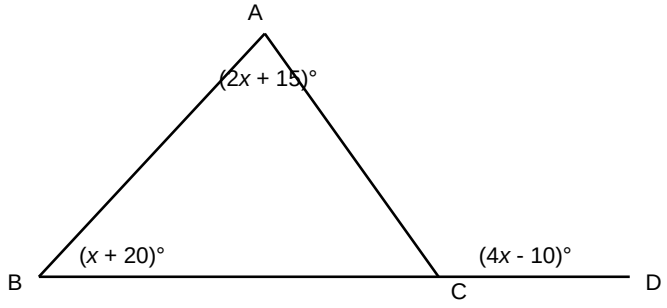


Diagram **NOT** accurately drawn

The diagram shows triangle ABC with the side BC extended to D. Work out the size of angle ACD.

4.3

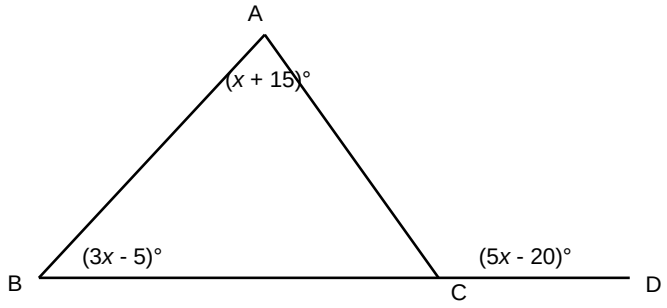


Diagram **NOT** accurately drawn

The diagram shows triangle ABC with the side BC extended to D. Work out the size of angle ACD.

Practice: the exterior angle of a triangle (continued)

4.4

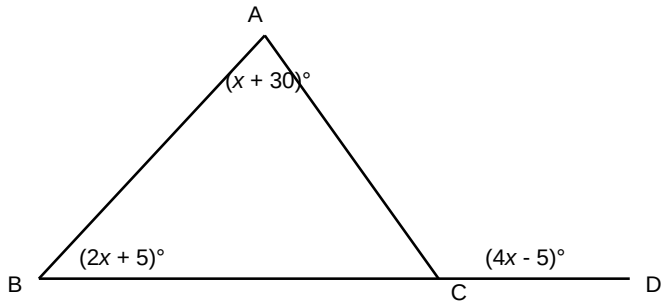


Diagram **NOT** accurately drawn

The diagram shows triangle ABC with the side BC extended to D.
Work out the size of angle ACD.

4.5

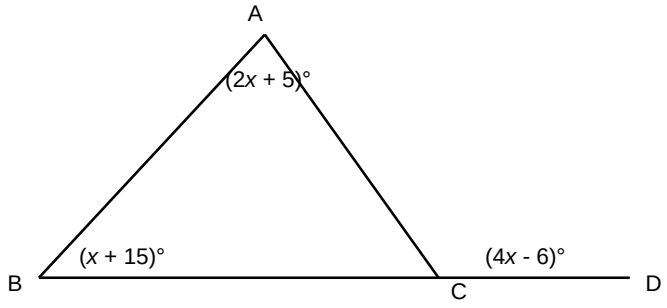


Diagram **NOT** accurately drawn

The diagram shows triangle ABC with the side BC extended to D.
Work out the size of angle ACD.

4.6

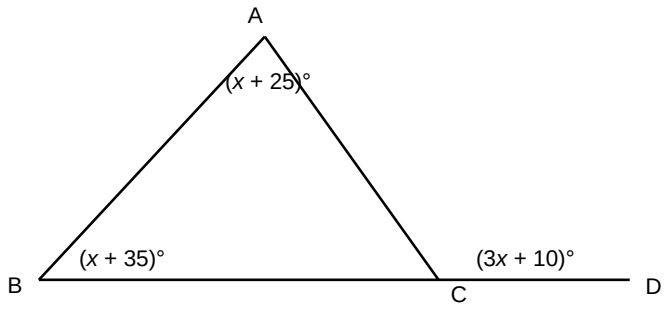


Diagram **NOT** accurately drawn

The diagram shows triangle ABC with the side BC extended to D.
Work out the size of angle ACD.

Shape and Space 1 · Angles (Lessons 5.1, 5.3)

Worked example: angles in a parallelogram

$ABCD$ is a parallelogram. Angle $A = (2x + 20)^\circ$ and angle $D = (3x + 10)^\circ$. Work out the size of the smallest angle.

Step 1	Co-interior angles add to 180°	$2x + 20 + 3x + 10 = 180$
Step 2	Simplify	$5x + 30 = 180$
Step 3	Solve	$x = 30$
Step 4	Angle A	$2 \times 30 + 20 = 80^\circ$
Step 5	Angle D	$3 \times 30 + 10 = 100^\circ$
Step 6	Answer	80°

Practice: angles in a parallelogram

4.7

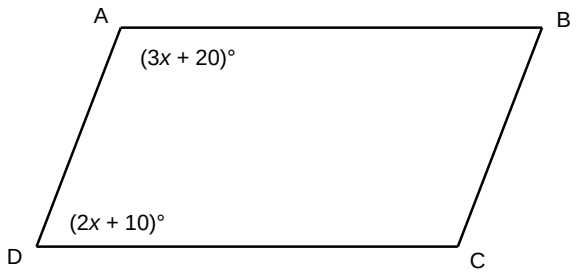


Diagram **NOT** accurately drawn

$ABCD$ is a parallelogram.
Work out the size of the smallest angle.

4.8

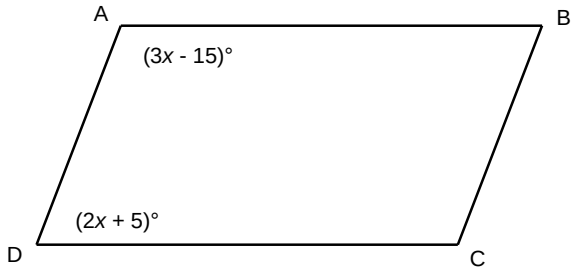


Diagram **NOT** accurately drawn

$ABCD$ is a parallelogram.
Work out the size of the smallest angle.

4.9

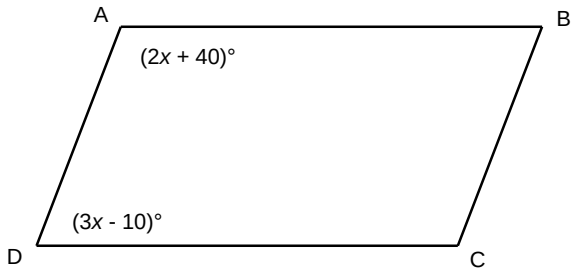


Diagram **NOT** accurately drawn

$ABCD$ is a parallelogram.
Work out the size of the smallest angle.

Practice: angles in a parallelogram (continued)

4.10

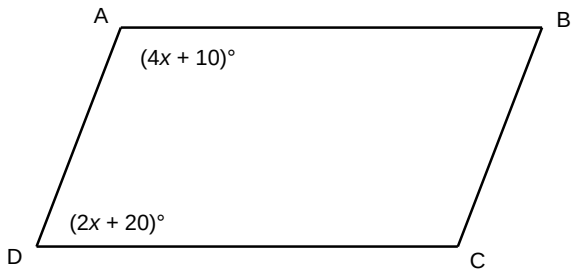


Diagram **NOT** accurately drawn

$ABCD$ is a parallelogram.
Work out the size of the smallest angle.

4.11

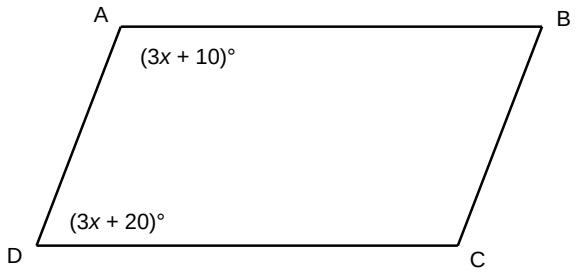


Diagram **NOT** accurately drawn

$ABCD$ is a parallelogram.
Work out the size of the smallest angle.

4.12

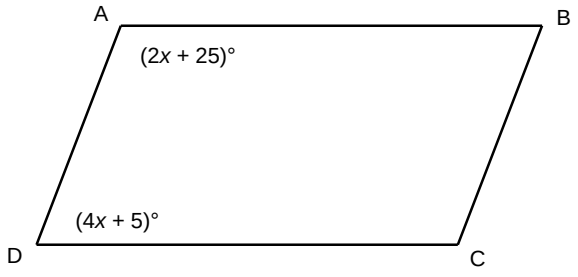


Diagram **NOT** accurately drawn

$ABCD$ is a parallelogram.
Work out the size of the smallest angle.

Shape and Space 1 · Pythagoras' theorem, area and circles (Lessons 5.2, 5.4, 5.5)

Worked example: finding a shorter side

A right-angled triangle has hypotenuse 12.5 cm. One shorter side is 7.2 cm. Work out the length x of the other side. Give your answer correct to 3 significant figures.

Step 1	Square and subtract	$x^2 = 12.5^2 - 7.2^2 = 104.41$
Step 2	Square root	$x = \sqrt{104.41} = 10.2181\ldots$
Step 3	Answer	$10.2\ \text{cm}$

Practice: finding a shorter side

5.1

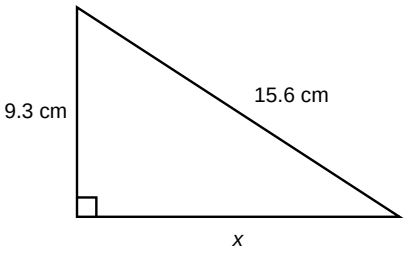


Diagram **NOT** accurately drawn

Work out the length x.
Give your answer correct to 3 significant figures.

5.2

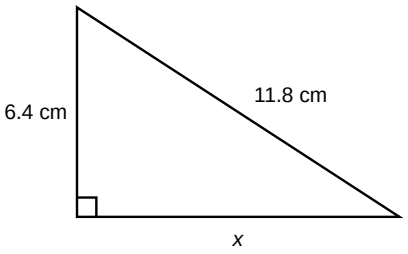


Diagram **NOT** accurately drawn

Work out the length x.
Give your answer correct to 3 significant figures.

5.3

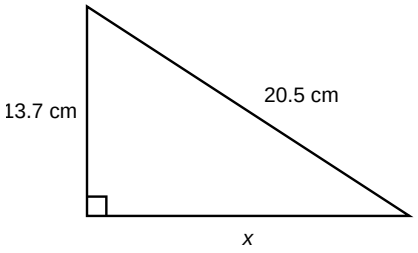


Diagram **NOT** accurately drawn

Work out the length x.
Give your answer correct to 3 significant figures.

Practice: finding a shorter side (continued)

5.4

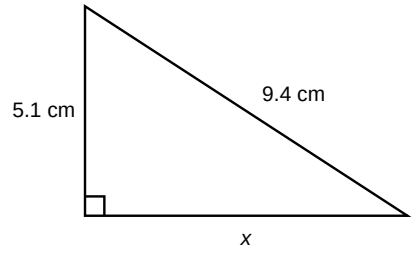


Diagram **NOT** accurately drawn

Work out the length x.
Give your answer correct to 3 significant figures.

5.5

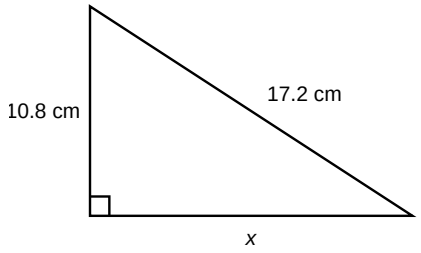


Diagram **NOT** accurately drawn

Work out the length x.
Give your answer correct to 3 significant figures.

5.6

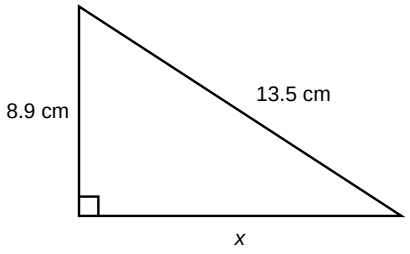


Diagram **NOT** accurately drawn

Work out the length x.
Give your answer correct to 3 significant figures.

Shape and Space 1 · Pythagoras' theorem, area and circles (Lessons 5.2, 5.4, 5.5)

Worked example: the area of a trapezium

A trapezium has parallel sides x cm and $(x + 2)$ cm. Its height is 5 cm. The area of the trapezium is 45 cm². Work out the value of x.

Step 1	Area formula	$\frac{1}{2}(x + x + 2) \times 5 = 45$
Step 2	Multiply by 2	$(2x + 2) \times 5 = 90$
Step 3	Divide by 5	$2x + 2 = 18$
Step 4	Solve	$x = 8$

Practice: the area of a trapezium

5.7

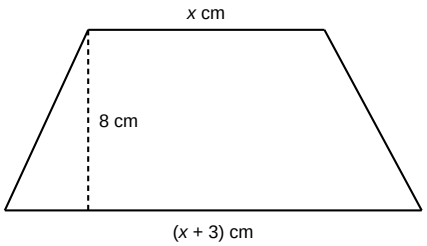


Diagram **NOT** accurately drawn

The area of the trapezium is 60 cm²
Work out the value of x.

5.8

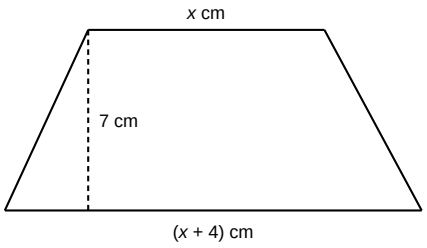


Diagram **NOT** accurately drawn

The area of the trapezium is 98 cm²
Work out the value of x.

5.9

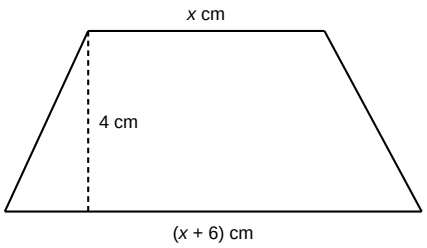


Diagram **NOT** accurately drawn

The area of the trapezium is 48 cm²
Work out the value of x.

Practice: the area of a trapezium (continued)

5.10

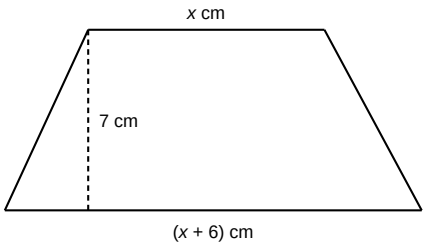


Diagram **NOT** accurately drawn

The area of the trapezium is 91 cm²
Work out the value of x.

5.11

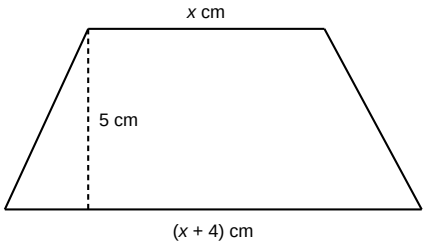


Diagram **NOT** accurately drawn

The area of the trapezium is 55 cm²
Work out the value of x.

5.12

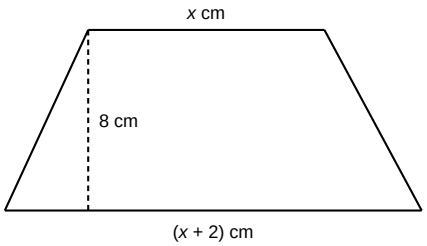


Diagram **NOT** accurately drawn

The area of the trapezium is 120 cm²
Work out the value of x.

Shape and Space 1 · Pythagoras' theorem, area and circles (Lessons 5.2, 5.4, 5.5)

Worked example: radius from the area

A circle has area 200 cm². Work out the radius of the circle. Give your answer correct to 3 significant figures.

Step 1	$A = \pi r^2$	$\pi r^2 = 200$
Step 2	Divide by π	$r^2 = 200 \div \pi = 63.662\ldots$
Step 3	Square root	$r = 7.97885\ldots$
Step 4	Answer	$7.98\ \text{cm}$

Practice: radius from the area

5.13 A circle has area 120 cm². Work out the radius of the circle. Give your answer correct to 3 significant figures.

5.14 A circle has area 300 cm². Work out the radius of the circle. Give your answer correct to 3 significant figures.

5.15 A circle has area 85 cm². Work out the radius of the circle. Give your answer correct to 3 significant figures.

5.16 A circle has area 60 cm². Work out the radius of the circle. Give your answer correct to 3 significant figures.

5.17 A circle has area 250 cm². Work out the radius of the circle. Give your answer correct to 3 significant figures.

5.18 A circle has area 45 cm². Work out the radius of the circle. Give your answer correct to 3 significant figures.